



Attention-guided multi-feature fusion hybrid architecture for person re-identification in smart surveillance systems

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ABSTRACT

Person re-identification (PReID) remains challenging due to illumination changes, occlusions, background clutter, and resolution degradation that obscure identity cues. Conventional CNN-based methods often struggle to progressively reinforce discriminative information across hierarchical feature levels, particularly when attention is applied only at selected stages, while transformer-based approaches introduce high computational cost. To address these limitations, we propose FocusReID, a lightweight PReID framework that integrates a Fusion Attention (FA) module into a DenseNet-121 backbone. The proposed multi-stage attention design enables progressive hierarchical refinement of feature representations, strengthening identity-discriminative cues with minimal additional complexity. FocusReID further employs warm-up training and a hybrid loss combining Circle Loss and Triplet Loss to improve optimization stability and retrieval-oriented feature learning. Extensive experiments demonstrate that FocusReID achieves competitive performance, reaching 95.71% Rank-1 accuracy on Market-1501 and 91.10% on DukeMTMC-reID, while operating 6.8× faster than transformer-based models such as SwinTrans-L. Additional ablation, cross-dataset, and visualization analyses confirm the effectiveness and robustness of the proposed design.

1. Introduction

Person re-identification (PReID) is a fundamental task in computer vision that aims to identify individuals across non-overlapping camera views, as illustrated in Fig. 1. It has important applications in surveillance, security, and intelligent monitoring systems [1,2]. Despite the remarkable progress achieved through deep learning (DL), PReID remains challenging due to substantial appearance variations caused by changes in pose, illumination, occlusion, scale, and background clutter [3], as illustrated in Fig. 2. These challenges require robust feature representations capable of capturing discriminative identity information in complex environments [4,5].

Deep learning, particularly Convolutional Neural Networks (CNNs), has played a central role in advancing PReID by enabling hierarchical feature extraction. Several neural network architectures, including ResNet [6], EfficientNet [7], and DenseNet [8], have demonstrated strong performance in PReID tasks. Moreover, conventional CNN-based PReID models often face difficulties in preserving and refining identity-discriminative representations across hierarchical layers. Although localized receptive fields and repeated downsampling operations can weaken fine-grained spatial details [9], an additional challenge lies in

how informative representations are progressively developed throughout the network. In many existing studies, attention mechanisms are commonly incorporated as plug-in modules or applied at selected stages of the backbone, rather than being explicitly examined as a means of continuous refinement across multiple hierarchical levels. Such designs may limit the extent to which discriminative cues are reinforced during intermediate feature propagation, where identity-relevant information is still being formed. Consequently, subtle appearance characteristics, local textures, and part-aware cues may not always be sufficiently emphasized under challenging conditions such as occlusion, background clutter, and large appearance variations. Although transformer-based models improve global context modeling [10], their computational and memory requirements may reduce their practicality in real-world surveillance settings [11].

In the proposed FocusReID framework, DenseNet-121 (D121) is employed as the feature extraction backbone because its dense connectivity encourages feature reuse and supports stable gradient propagation [8]. However, although D121 provides an effective foundation for representation learning, it does not explicitly model attention to identity-relevant regions or channels. As a result, informative

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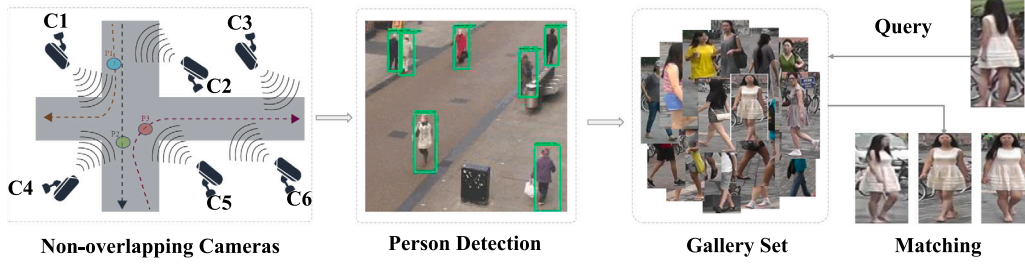


Fig. 1. An example of PREID pipeline: (a) Image capture from non-overlapping cameras C1–C6, (b) person detection, (c) gallery construction, and (d) query matching for cross-camera identification.



Fig. 2. Common challenges in PREID, including: (a) low resolution, (b) scale variation, (c) occlusion, (d) illumination, (e) camera viewpoint, and (f) domain gap.

cues may not be sufficiently emphasized under challenging conditions such as occlusion, background clutter, and large appearance variations. To address this issue, FocusReID incorporates a lightweight attention mechanism across multiple dense blocks to encourage hierarchical refinement of identity-relevant features while maintaining computational efficiency. Moreover, existing attention-enhanced CNN-based PREID methods often apply attention as standalone modules or only at selected backbone stages, which may limit the continuous refinement of identity-discriminative features across hierarchical layers. In contrast, FocusReID integrates lightweight attention after each dense block, enabling progressive multi-stage refinement of feature representations. Although the FA module follows the design principle of channel and spatial attention similar to CBAM, our contribution lies in its systematic integration across the DenseNet backbone to support continuous hierarchical refinement. As further validated in our ablation study (Table 7), progressively integrating FA across multiple dense blocks leads to consistent performance improvements, highlighting the effectiveness of the proposed hierarchical refinement.

To address these limitations, we propose an efficient PREID architecture with an integrated global-local feature enhancement module, termed Fusion Attention (FA), based on the design principle of the CBAM framework [12]. Although the FA module follows the paradigm of sequential channel and spatial attention, the main contribution of this work lies in its systematic multi-stage integration within the backbone to enable progressive hierarchical refinement for PREID. The FA module captures global-local feature interactions by combining Channel Attention (CA) and Spatial Attention (SA), where CA performs global feature re-weighting and SA focuses on local region refinement. This design strengthens the network’s ability to emphasize discriminative identity cues while suppressing irrelevant background information,

thereby improving robustness under occlusion and complex visual conditions [13]. To further improve generalization, we employ Random Erasing (R-E) [14] and a warm-up strategy (W-up) to stabilize optimization. In addition, Circle Loss (CL) and Triplet Loss (TL) [15] are used to enhance embedding discriminability. Experiments on benchmark PREID datasets show that FocusReID achieves competitive performance on Market-1501 and DukeMTMC-reID, demonstrating the effectiveness of progressive attention integration within a densely connected backbone. The key contributions of this study are summarized as follows.

- We present *FocusReID*, a lightweight PREID framework based on D121 and enhanced with a Fusion Attention (FA) module for progressive multi-level feature refinement.
- We introduce a multi-stage attention integration strategy that enables progressive hierarchical refinement of feature representations, improving robustness under occlusion, low resolution, and background clutter while maintaining computational efficiency.
- We conduct extensive experiments, including within-dataset evaluation, cross-dataset generalization, ablation studies, and efficiency analysis. The results indicate that FocusReID achieves competitive retrieval performance with a favorable accuracy–efficiency trade-off. In addition, Grad-CAM visualizations highlight improved focus on identity-relevant body regions, while comparisons on Market-1501 across DenseNet, ResNet, ConvNeXt, HRNet, EfficientNet-B4, and transformer-based backbones further support the effectiveness of the proposed design.

The remainder of this paper is organized as follows. Section 2 reviews related work, Section 3 introduces FocusReID, Section 4–4.3

present the experimental setup and comparison results, Section 5 provides ablation studies, and Section 6 concludes the paper.

2. Literature review

This section explores two key research areas: fundamental PReID methods and attention mechanisms, highlighting their development, interplay, and remaining challenges in feature representation.

2.1. Fundamental techniques

PReID is an important computer vision task with applications in surveillance, security, and human–computer interaction. Early approaches relied on handcrafted descriptors such as color histograms and texture features [16]. Although these methods provided basic appearance cues, they were highly sensitive to variations in pose, illumination, occlusion, and background clutter. Deep learning, particularly convolutional neural networks (CNNs), transformed PReID by enabling automatic extraction of discriminative representations. Siamese architectures and triplet-loss formulations [15] improved cross-view matching by learning feature similarity directly. More recent work has explored transformer-based and reconstruction-oriented strategies for challenging scenarios such as occluded PReID, as exemplified by FRCE [17].

Recent developments have focused on part-based modeling, occlusion handling, and efficient representation learning. Sarfraz et al. [18] addressed pose misalignment through keypoint-guided feature extraction, while resolution degradation has been mitigated using pose-driven super-resolution [19] and ESRGAN-based enhancement [20]. Additional advances include random erasing for occlusion robustness [14], spatial–temporal attention for video-based PReID [21], and transformer models employing self-attention to capture long-range dependencies [22]. Arain et al. [10] demonstrated that integrating channel attention into Swin Transformer architectures supports strong patch-level representation learning. Broader extensions incorporate cross-modality interaction, hierarchical modeling, and synthetic data augmentation. Zou and Chen [23] introduced Modality Interactive Attention for visible–infrared alignment, and Lu et al. [24] developed a dual-branch adaptive attention Transformer for occluded person retrieval. Zhuang et al. [4] further improved foreground–background separation through a Feature Attention Module. Multimodal and hierarchical frameworks have also shown promise: Hong et al. [5] proposed UHDNet with hierarchical dependency learning for modality harmonization, while Zuo et al. [3] leveraged spatio-temporal attention with Transformers for video-based VI-ReID. Delussu et al. [25] demonstrated that well-designed synthetic datasets can significantly improve model generalization.

More recently, PReID has advanced through multi-source domain adaptation, generative data expansion, and trajectory-aware modeling. Xian et al. [26] improve cross-domain consistency via multi-source expert distillation, Siddiqui et al. [27] use diffusion-based data expansion to handle clothing variation, and Zhang et al. [28] explore temporal embedding for cross-camera retrieval. Although effective, these methods often require additional data, auxiliary supervision, or more complex training pipelines. Despite recent progress, PReID remains challenging under occlusion, background clutter, and large appearance variations. CNN-based methods often struggle to capture long-range dependencies, while transformer-based models improve global context modeling at the cost of higher computational overhead. These limitations motivate efficient architecture-level solutions that enhance discriminative feature learning without increasing model complexity.

2.2. Attention mechanisms

Attention mechanisms are crucial in PReID, enabling the effective mitigation of issues such as background clutter and occlusions. They achieve this by focusing on discriminative regions and channels. Traditional deep learning models often struggle with such challenges due to their limited receptive fields. The SA helps localize relevant regions, as demonstrated by Li et al.’s Harmonious Attention Network [29]. The CA, on the other hand, emphasizes the importance of global features, as seen in SENet [30] and CBAM [12], with the latter combining both SA and CA to achieve more refined feature representations. Further advancements, such as Horizontal Pyramid Matching (HPM) [31], enhance the discriminative learning capability of attention-based models. Recent work, Transformer-based methods [32,33] introduce self-attention mechanisms to capture long-range dependencies, though often at the cost of increased computational complexity. SwinReID [10] demonstrates the advantages of combining hierarchical transformers with channel attention for more effective feature extraction. CBAM [12] stands out due to its dual attention mechanism, and its effectiveness has been validated in multiple studies [31]. The FocusReID framework further advances this area by enhancing the Densely Connected Convolutional Network D121 through the integration of an FA module. While D121 excels at feature reuse, it lacks the ability to prioritize spatial and channel information. The FA module addresses this limitation by incorporating both SA and CA to suppress background noise, handle occlusions, highlight key body regions, and balance computational cost and complexity. As a result, FocusReID significantly improves performance under conditions of extreme appearance variation.

3. FocusReID architecture

In this section, we present FocusReID, a novel PReID approach that integrates a dual attention FA module into the D121 backbone. It is further enhanced with RE, LW-up, and advanced loss functions. FocusReID comprises two main components: feature extraction and attention-based refinement.

The FA module, comprising CA and SA, improves the learning of informative features while suppressing irrelevant ones, thus enhancing representation learning. The final classification is performed through a fully connected (FC) layer. The architecture, illustrated in Fig. 3, emphasizes the integration of attention mechanisms and training enhancements, resulting in improved discriminative feature learning for PReID tasks.

The network begins by taking an input image $X \in \mathbb{R}^{3 \times H \times W}$, which is first processed by a 7×7 convolutional layer, followed by batch normalization, ReLU activation, and max pooling. The extracted low-level features are then passed through four dense blocks, denoted as $D_1[6]$, $D_2[12]$, $D_3[24]$, and $D_4[16]$, interleaved with three transition layers T_1 , T_2 , and T_3 . Each transition layer includes a 1×1 convolution followed by a 2×2 average pooling operation, which helps to reduce the spatial dimensions and control the complexity of the network. The detailed architecture and flow of the network are illustrated in Algorithm 1.

To enhance the feature learning process, an FA module is introduced after each dense block. This component combines CA and SA mechanisms, which allows the network to focus on more relevant and discriminative features. The attention refinement is formulated as:

$$FA_i(F) = CA(F) \odot SA(F) \quad (1)$$

where $CA(F)$ and $SA(F)$ represent the channel and SA maps respectively, and $\odot$ denotes element-wise multiplication.

The CA mechanism focuses on important feature channels, while the SA mechanism identifies salient spatial regions. Together, they adaptively refine the feature map.

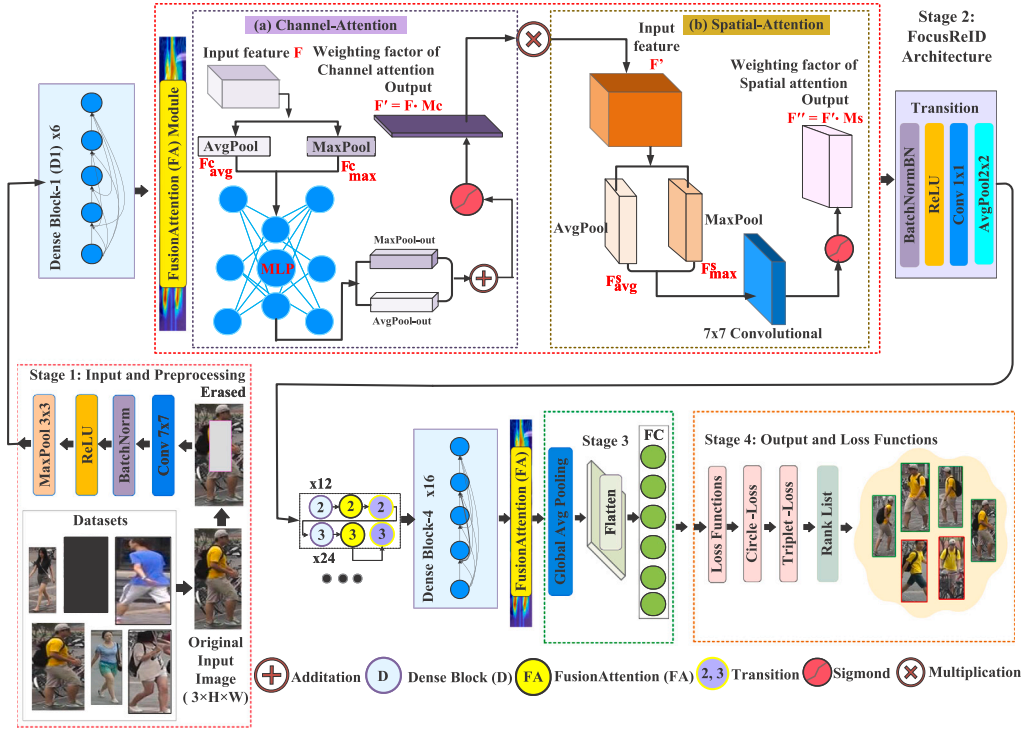


Fig. 3. The architecture of the proposed FocusReID framework.

The complete attention-enhanced feature extraction process can thus be expressed as:

$$F_{\text{final}} = A_{\text{FA}}^4 \left(D_4^{[16]} \circ T_3 \circ A_{\text{FA}}^3 \left(D_3^{[24]} \circ T_2 \circ A_{\text{FA}}^2 \left(D_2^{[12]} \circ T_1 \circ A_{\text{FA}}^1 \left(D_1^{[6]}(F_{\text{in}}) \right) \right) \right) \right) \quad (2)$$

where $A_{\text{FA}}^i(\cdot)$ denotes the application of the FA module after the i th dense block. The use of $\circ$ indicates the sequential composition of layers. This hierarchical integration of attention mechanisms enables the model to capture more discriminative and robust representations by dynamically adjusting both spatial and channel-wise feature importance at multiple levels of abstraction.

The complete mathematical model describing the execution flow of FocusReID is formulated as follows:

Backbone Structure, Transition Layers, and FA Modules. The architecture comprises DenseNet blocks $D_1^{[6]}, D_2^{[12]}, D_3^{[24]}, D_4^{[16]}$, transition layers

$$T_i = P_i \circ C_i^{1 \times 1}, \quad i \in \{1, 2, 3\},$$

and optional FA modules $A_{\text{FA}}^1 - A_{\text{FA}}^4$ inserted after the dense blocks.

The execution flow is given by:

$$\begin{aligned} F_0 &= \text{Conv0}(X), \\ F_1 &= D_1^{[6]}(F_0), \quad \hat{F}_1 = A_{\text{FA}}^1(F_1) \text{ or } F_1, \quad F_1' = T_1(\hat{F}_1), \\ F_2 &= D_2^{[12]}(F_1'), \quad \hat{F}_2 = A_{\text{FA}}^2(F_2) \text{ or } F_2, \quad F_2' = T_2(\hat{F}_2), \\ F_3 &= D_3^{[24]}(F_2'), \quad \hat{F}_3 = A_{\text{FA}}^3(F_3) \text{ or } F_3, \quad F_3' = T_3(\hat{F}_3), \\ F_4 &= D_4^{[16]}(F_3'), \quad \hat{F}_4 = A_{\text{FA}}^4(F_4) \text{ or } F_4, \\ F_{\text{final}} &= \text{Norm5}(\hat{F}_4). \end{aligned}$$

The final representation is obtained as

$$f = \text{Flatten}(\text{GAP}(F_{\text{final}})),$$

which is used for embedding learning and person re-identification.

Algorithm 1 Proposed FocusReID Network

Require: Person image $X \in \mathbb{R}^{3 \times H \times W}$, label y

Ensure: Feature vector $\mathbf{f} \in \mathbb{R}^d$, loss L

- 1: **Input Preprocessing:**
- 2: $\tilde{X} \leftarrow \text{RandomErasing}(X, p)$ $\triangleright$ Augmentation by randomly occluding a region
- 3: **Hierarchical Feature Extraction:**
- 4: $F_{\text{in}} \leftarrow \tilde{X}$
- 5: **for** each dense block $D_i \in \{D_1^{[6]}, D_2^{[12]}, D_3^{[24]}, D_4^{[16]}\}$ **do**
- 6: $F \leftarrow D_i(F_{\text{in}})$
- 7: **Fusion Attention:**
- 8: *Channel Attention:*
- 9: $M_c \leftarrow \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F)))$
- 10: $F' \leftarrow M_c \odot F$
- 11: *Spatial Attention:*
- 12: $M_s \leftarrow \sigma(\text{Conv}_{7 \times 7}[\text{AvgPool}(F'); \text{MaxPool}(F')])$
- 13: $F \leftarrow M_s \odot F'$
- 14: **if** $D_i \neq D_4^{[16]}$ **then**
- 15: $F_{\text{in}} \leftarrow T_i(F)$ $\triangleright$ Apply transition (pooling/conv)
- 16: **end if**
- 17: **end for**
- 18: **Classification and Loss Computation:**
- 19: $\mathbf{f} \leftarrow \text{GAP}(F)$
- 20: $\mathbf{p} \leftarrow \text{FC}(\mathbf{f})$
- 21: $L_{\text{circle}} \leftarrow \text{CircleLoss}(\mathbf{f}, y)$
- 22: $L_{\text{triplet}} \leftarrow \text{TripletLoss}(\mathbf{f}, y)$
- 23: $L \leftarrow \lambda_1 L_{\text{circle}} + \lambda_2 L_{\text{triplet}}$
- 24: **Ranking and Evaluation:**
- 25: Compute similarity between $\mathbf{f}$ and gallery features
- 26: Evaluate using mAP and CMC metrics
- 27: **return** $\mathbf{f}, L$

3.1. Fusion attention module

In order to bolster the representational fidelity and class separability of extracted features, the FA module is incorporated into the pipeline and operates on the feature maps F_{out} . The FA module sequentially employs CA and SA to modulate feature importance across channel-wise and spatial dimensions. A visual representation of this dual-attention framework is provided in Fig. 4, and its operational mechanics are detailed below.

Channel attention module

The channel attention mechanism computes channel-wise weights to emphasize informative features and suppress less relevant ones. This is achieved through global average pooling and max pooling operations:

$$F_{\text{avg}}^c = \text{AvgPool}(F_{\text{out}}), \quad F_{\text{max}}^c = \text{MaxPool}(F_{\text{out}}). \quad (3)$$

The pooled representations are then passed through a shared multi-layer perceptron (MLP), and sigmoid activation is applied to each branch separately. The resulting channel attention weights are computed as:

$$M_c = \sigma(\text{MLP}(F_{\text{avg}}^c)) + \sigma(\text{MLP}(F_{\text{max}}^c)). \quad (4)$$

where σ denotes the sigmoid activation function, and M_c represents the channel attention weights. The refined feature map is then obtained as:

$$F' = M_c \odot F_{\text{out}}. \quad (5)$$

Spatial attention module

The spatial attention (SA) mechanism focuses on identifying salient regions within the feature maps. It computes spatial attention weights by applying average pooling and max pooling along the channel dimension:

$$F_{\text{avg}}^s = \text{AvgPool}(F'), \quad F_{\text{max}}^s = \text{MaxPool}(F'). \quad (6)$$

The pooled feature maps are concatenated along the channel axis and passed through a 7×7 convolutional layer followed by a sigmoid activation:

$$M_s = \sigma(\text{Conv}_{7 \times 7}([F_{\text{avg}}^s; F_{\text{max}}^s])). \quad (7)$$

The final refined feature representation is computed as:

$$F'' = M_s \odot F'. \quad (8)$$

Feature pooling and classification

The refined feature maps F'' are processed using global average pooling (GAP) to generate a compact feature vector:

$$F_{\text{global}} = \text{GAP}(F''). \quad (9)$$

This pooled feature vector is passed through a fully connected (FC) layer for classification:

$$L = \text{FC}(F_{\text{global}}). \quad (10)$$

where L represents the final output logits. The model is optimized using a combination of Circle Loss (CL) and Triplet Loss (TL) to ensure robust feature representation and discriminative capability.

3.2. Training optimization

To improve generalization and robustness, we apply R-E, LRW-up, and a hybrid of CL and TL losses. These strategies enhance feature discriminability, inter-class separation, and representation quality. Experiments show that combining CL and TL yields superior performance by effectively balancing classification and metric learning.

Random-Erasing (R-E) improves model robustness by randomly masking image regions during training [14]. The augmented image $\tilde{X}$ is defined as:

$$\tilde{X} = \begin{cases} X_{\text{occl}} & \text{with probability } \gamma, \\ X & \text{otherwise,} \end{cases} \quad (11)$$

where X is the original image and X_{occl} has a randomly erased patch. Object-aware variants mask both foreground and background areas, as shown in Fig. 5.

Warm-up Strategy (W-up): To ensure stable optimization at the beginning of training, the learning rate α is increased linearly during the first T steps:

$$\alpha = \alpha_0 + \left(\frac{\alpha_1 - \alpha_0}{T} \right) \cdot t, \quad (12)$$

where α_0 and α_1 denote the starting and maximum learning rates, T is the warm-up period, and t is the current training step or epoch.

Circle Loss (CL): CL emphasizes harder positive and negative pairs to strengthen feature discriminability:

$$L_{\text{CL}} = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N [y_{ij} \max(0, m_1 + \Delta_i - \cos \theta_{ij}) + (1 - y_{ij}) \max(0, \cos \theta_{ij} - m_2)], \quad (13)$$

where $\cos(\theta_{ij})$ is the cosine similarity between feature vectors, y_{ij} is the pair label, and m_1, m_2 are margins.

Triplet Loss (TL): drives the network to minimize the distance between positive pairs while maximizing it for negative pairs:

$$L_{\text{TL}} = \max(0, d_{\text{pos}} - d_{\text{neg}} + m), \quad (14)$$

where d_{pos} and d_{neg} are the distances between the anchor and positive/negative samples, respectively, and m is the margin.

Total Loss Function: To effectively balance classification and metric learning objectives, we combine CL and TL into a unified loss function:

$$L_{\text{Total}} = \lambda_{\text{CL}} L_{\text{CL}} + \lambda_{\text{TL}} L_{\text{TL}}, \quad (15)$$

where λ_{CL} and λ_{TL} are weighting factors controlling the contributions of each loss function.

These techniques collectively boost the network's performance in challenging re-identification tasks. The combination of CL and TL outperforms other configurations, ensuring a robust and accurate solution for PReID.

4. Experimental setup

This section presents FocusReID's evaluation on benchmark datasets, outlines the implementation details, compares it with representative state-of-the-art methods, and analyzes its components through ablation studies and case analysis.

4.1. PReID datasets

An overview of the datasets utilized in our experiments is provided in this section, namely Market-1501, DukeMTMC-reID, and Occluded-Duke. A statistical summary is presented in Table 1. We validate FocusReID on four widely used PReID benchmarks. **Market-1501** [34] comprises 32,668 images of 1501 unique identities captured across six cameras. Its training split contains 12,936 images of 751 subjects, while

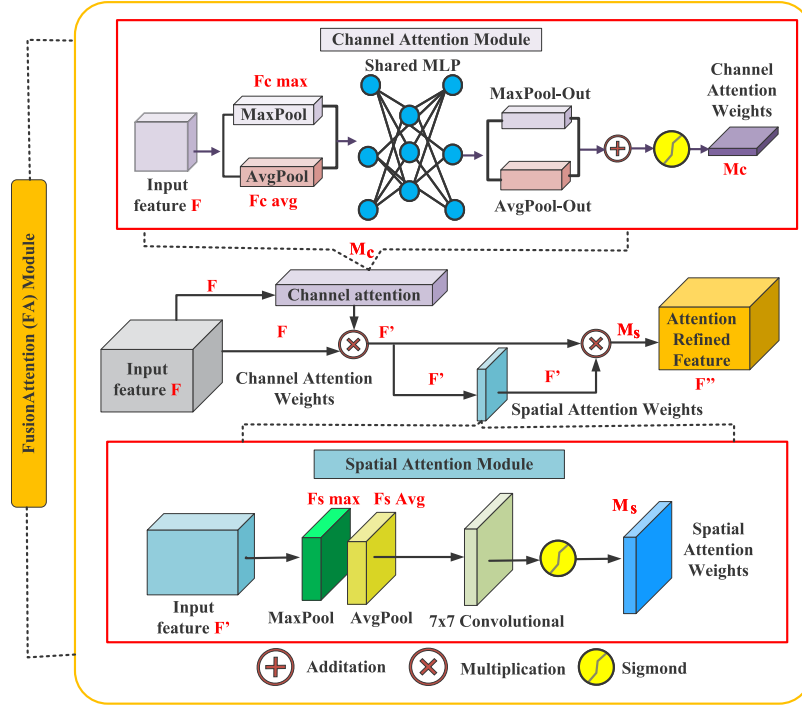


Fig. 4. FA mechanism architecture. The input feature F is refined by SA and CA modules. SA weights use max/average pooling and convolution; CA weights use max/average pooling and a shared MLP. The refined feature F'' combines these attentions.

Table 1

PRiD datasets list identities and image counts for training (T), query (Q), and gallery (G) sets.

Datasets	ID	T	Q	G
DukeMTMC-reID	1404	16,522	2228	17,661
Occluded-Duke	1404	15,618	2210	17,661
Market-1501	1501	12,936	3368	16,364

the testing set holds 19,732 images corresponding to 750 individuals. The query subset includes 3368 images. **DukeMTMC-reID** [35] is a more challenging dataset featuring 36,411 images of 1404 identities plus 408 distractor images, collected via eight cameras. The training data consists of 16,522 images of 702 persons; the test gallery has 17,661 images of 702 individuals along with 408 distractors. Its query set includes 2228 images. Finally, **Occluded-Duke** [36] addresses occlusion scenarios, containing 15,618 training images from 702 identities, a gallery of 17,661 images, and 2210 occluded query images of 1110 persons.

4.2. Implementation details

This section describes the training setup, backbone model, loss functions, and optimization strategy of FocusReID. We use D121, initialized with ImageNet-22K pre-trained weights, as the backbone and remove its classification head for feature extraction. The FA module is integrated to enhance feature discrimination through adaptive channel-wise and spatial refinement. For evaluation, datasets are used at their original resolutions, while Market-1501 is additionally downsampled from 64×128 to 32×64 and 16×32 to assess robustness under low-resolution settings.

Extensive experiments evaluate several loss functions, including Cross-Entropy Loss (CEL), ArcFace Loss (AFL), Circle Loss (CL), CosFace Loss (CFL), SphereFace Loss (SFL), Triplet Loss (TL), Contrastive Loss (CTL), and Ring Loss (RLL). Cross-Entropy Loss (CEL), also known as Identity Loss, is included as a standard classification-based baseline

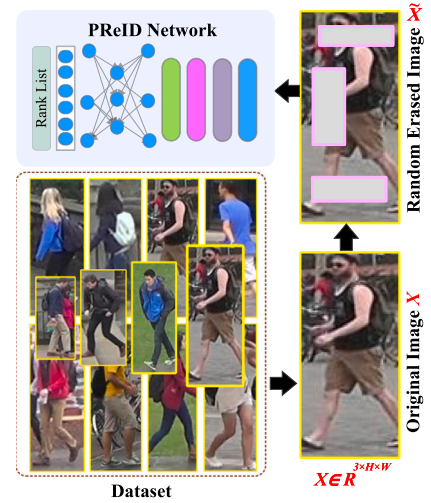


Fig. 5. Illustration of Random Erasing (R-E): a region of the original image X is randomly occluded, producing $\tilde{X}$.

in PRiD. However, Table 9 shows that the combination of Circle Loss (CL) and Triplet Loss (TL) achieves the best overall performance. While CEL mainly learns class-level decision boundaries, CL and TL directly optimize the embedding space by reducing intra-class variation and enlarging inter-class separation, which is more suitable for retrieval-based ReID tasks. Therefore, CL and TL are selected as the final loss configuration based on experimental comparison rather than by excluding Identity Loss a priori. Although CEL remains an effective supervision strategy for identity classification, the CL+TL combination proves more suitable for retrieval-oriented embedding learning in our setting. Training uses SGD with batch size 32, learning rate $\alpha = 0.02$, and weight decay 0.0005. A 5-epoch linear warm-up stabilizes convergence. Dropout (0.5) and Random Erasing (0.5) are used to mitigate overfitting. The model is trained for 200 epochs on an NVIDIA

Table 2
Performance comparison on Market-1501 dataset.

Method	R1 (%)	R5 (%)	mAP (%)
CNN-based			
Deep-Person [2]	92.30%	–	79.60%
MHSA-Net [6]	94.60%	–	84.00%
DRL-Net [†] [9]	94.70%	–	86.90%
FED [†] [39]	95.00%	–	86.30%
DCAL [40]	94.70%	–	87.50%
SCS+ [41]	90.30%	–	89.40%
QAConv+GS [42]	91.60%	–	75.50%
JAD+RK [43]	91.00%	–	87.50%
SCAT [13]	95.10%	–	88.00%
BPBreIDRN [16]	95.10%	–	87.10%
GPEOG [32]	94.80%	–	84.50%
RTGAT [44]	95.30%	–	88.20%
FRCE [17]	95.20%	–	88.10%
FPA-Net [37]	96.10%	–	<u>89.60%</u>
Transformer-based			
DeiT-B/16 [38]	94.40%	–	86.60%
ViT-B/16 [38]	94.70%	–	86.80%
DenseFormer [11]	95.10%	–	88.10%
DRL-Net [†] [9]	94.70%	–	86.90%
FED [†] [39]	95.00%	–	86.30%
CLIP-ReID [22]	95.40%	–	90.50%
MTIPE [45]	95.20%	–	89.00%
MDFA [1]	95.80%	–	91.40%
FocusReID (Ours)	<u>95.71%</u>	96.67%	91.95%

Note: The best results are in **bold**, and the second-best are underlined.

Tesla V100 GPU with data augmentation including 90° rotations and horizontal flips.

4.3. FocusReID SOTA evaluation

A detailed evaluation of the FocusReID model on three widely recognized PReID benchmarks is presented in this section. DukeMTMC-reID, Occluded-Duke, and Market-1501. Evaluation uses mean Average Precision (mAP) and Cumulative Matching Characteristic (CMC) at top-K ranks (R). FocusReID's performance is compared with recent SOTA methods. The results in **bold** represent the best scores; those underlined are the second best.

Market-1501: As shown in Table 2, FocusReID achieves competitive performance, reaching an mAP of **91.95%**, which is higher than the 89.60% reported by FPA-Net [37]. In terms of Rank-1 (R1) accuracy, FocusReID attains 95.71%, outperforming BPBreIDRN [16] and SCAT [13], both of which achieve 95.10%. Although FPA-Net yields the highest R1 score of 96.10%, FocusReID demonstrates superior retrieval strength with an R5 accuracy of 96.67%. When compared to transformer-based baselines, FocusReID consistently surpasses traditional variants such as DeiT-B/16 and ViT-B/16 [38], and outperforms more advanced models including Denseformer [11] (88.10% mAP) and CLIP-ReID (90.50%). These improvements underscore the advantages of FocusReID's attention-guided design and its ability to model contextual features more effectively.

DukeMTMC-reID: Results evaluation on the DukeMTMC PReID dataset is presented in Table 3. FocusReID achieves the highest mAP of **85.80%**, outperforming FRCE [17] (83.00%) and FPA-Net (81.90%). Its R1 accuracy of 91.10% is modestly lower than that of FRCE and FPA-Net (both at 91.30%), but remains highly competitive. FocusReID also records an R5 accuracy of 94.80%, further validating its retrieval capability. Compared to transformer-based methods such as Denseformer (81.31%) and CLIP-ReID (81.30%), FocusReID achieves an mAP improvement of 4–6 percentage points. This margin reflects the model's effective feature representation and robust generalization across diverse camera views and identity variations.

Occluded-Duke: Table 4 summarizes the evaluation on the Occluded-Duke dataset. FocusReID obtains the highest mAP of **63.42%**,

Table 3
Performance comparison on DukeMTMC-reID dataset.

Method	R1 (%)	R5 (%)	mAP (%)
CNN-based			
Deep-Person [2]	80.90%	–	64.80%
MHSA-Net [6]	87.30%	–	73.10%
CDNet [46]	88.60%	–	76.80%
DRL-Net [†] [9]	88.10%	–	76.60%
FED [†] [39]	89.40%	–	78.00%
JAD+RK [43]	82.70%	–	77.00%
SCS+ [41]	90.30%	–	80.90%
DCAL [40]	89.00%	–	80.10%
FED [39]	89.40%	–	78.00%
SCAT [13]	89.30%	–	79.80%
GPEOG [32]	87.50%	–	75.50%
RTGAT [44]	89.10%	–	80.20%
BPBreIDRN [16]	89.60%	–	78.30%
FRCE [17]	91.30%	–	<u>83.00%</u>
FPA-Net [37]	91.30%	–	81.90%
Transformer-based			
DeiT-B/16 [38]	89.30%	–	78.90%
ViT-B/16 [38]	88.80%	–	79.30%
DenseFormer [11]	90.10%	–	81.30%
DRL-Net [†] [9]	88.10%	–	76.60%
CLIP-ReID [22]	90.10%	–	81.30%
MTIPE [45]	90.00%	–	81.80%
MDFA [1]	91.20%	–	83.90%
FocusReID (Ours)	<u>91.10%</u>	94.80%	85.80%

Note: The best results are in **bold**, and the second-best are underlined.

Table 4
Performance comparison on Occluded-Duke dataset.

Method	R1 (%)	R5 (%)	mAP (%)
CNN-based			
HOREID [47]	55.10%	–	43.80%
ISP [48]	62.80%	–	52.30%
PVPM [49]	47.00%	–	37.70%
PAT [50]	64.50%	–	53.60%
MHSA-Net [6]	59.70%	–	44.80%
GPEOG [32]	64.10%	–	51.20%
BPBreIDRN [16]	66.70%	–	54.10%
SCAT [13]	62.80%	–	54.90%
Transformer-based			
DeiT-B/16 [38]	60.60%	–	53.10%
ViT-B/16 [38]	60.50%	–	53.10%
DenseFormer [11]	63.80%	–	55.60%
PAT [50]	64.50%	–	53.60%
MTIPE [45]	66.41%	–	57.82%
MDFA [1]	69.60%	–	62.40%
FocusReID (Ours)	<u>63.93%</u>	73.16%	63.42%

Note: The best results are in **bold**, and the second-best are underlined.

significantly outperforming CNN-based models such as BPBreIDRN (54.10%) and SCAT (54.90%), as well as transformer-based approaches including DeiT-B/16 (53.11%) and Denseformer (55.61%). Although its R1 accuracy of 63.93% is marginally lower than BPBreIDRN's **66.70%**, the higher mAP demonstrates a more consistent and comprehensive retrieval performance under occluded scenarios. FocusReID's R5 accuracy of 73.16% is competitive, though it trails SCAT (78.10%) and CLIP-ReID [22] (80.40%). These results highlight FocusReID's robustness against occlusions, pose variations, and lighting changes, enabled by its advanced attention-based architecture.

Across all three benchmark datasets, FocusReID consistently achieves strong performance, particularly in mAP, which better reflects retrieval robustness than R1 accuracy. While some methods slightly exceed FocusReID in Rank-1, its balanced results across metrics and datasets underscore its effectiveness in real-world scenarios. Its attention-guided feature extraction and robust representations enable it to handle occlusions, background clutter, and viewpoint variations. A summary of the results is shown in Fig. 6.

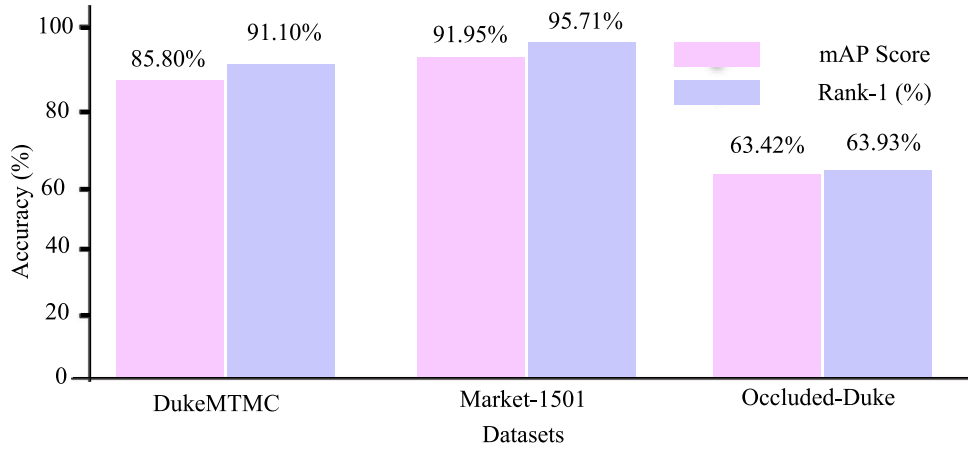


Fig. 6. mAP and R1 accuracy of FocusReID on benchmark datasets.

Table 5

Cross-dataset generalization performance of FocusReID.

Train → Test	R1 (%)	mAP (%)
Market → DukeMTMC	49.84%	30.33%
DukeMTMC → Market	52.01%	35.93%

Cross-Dataset Evaluation: Table 5 summarizes the cross-dataset evaluation results of FocusReID, where the model is trained on one benchmark dataset and tested on another. This setting provides a more challenging assessment of generalization by introducing domain shift between the training and testing data.

To further evaluate the generalization capability of FocusReID, we conduct cross-dataset experiments by training on one dataset and testing on another. As shown in Table 5, the proposed method achieves 52.01% R1 and 35.93% mAP for DukeMTMC-reID → Market-1501, and 49.84% R1 and 30.33% mAP for Market-1501 → DukeMTMC-reID. Although these results remain lower than within-dataset performance due to substantial domain shift, they indicate that the proposed FA-based hierarchical refinement provides reasonable transferability across datasets.

As shown in Table 5, FocusReID achieves reasonable retrieval performance under cross-dataset settings, indicating that the proposed framework can learn transferable identity representations. Although the performance is lower than that of within-dataset evaluation due to substantial domain shift, the results suggest that the FA-based hierarchical feature refinement contributes positively to generalization capability.

4.4. Parameter & computation analysis

We evaluate the FocusReID model against various backbone architectures, DenseNet (D121, D169, D201, D161), ResNet (ResNet50, ResNet101, ResNet152), ConvNeXt, HRNet, EfficientNet-B4 (Effi-B4), and Swin Transformers (SwinTrans_S, SwinTrans_L) on the PRiD Market-1501 dataset. Evaluation metrics include total trainable-parameters (Parms), multiply-accumulate operations (MACs), inference time (ITime) measured in seconds(s), GPU-memory usage, R1, and mAP, as shown in Table 6.

Compared to DenseNet models, FocusReID clearly outperforms all variants. Although D161 achieves the highest accuracy among them (92.61% R1 and 89.53% mAP), it demands significantly more resources: 5.12G MACs and 5210.12 MB GPU memory. FocusReID surpasses this with 95.71% R1 and 91.95% mAP, using only 1.89G MACs and 2715.12 MB GPU memory.

In the ResNet family, ResNet50 achieves 90.32% R1 with 2.70G MACs. ResNet101 and ResNet152 provide moderate gains in accuracy

(91.63% and 91.89% R1) but at a steep increase in computational cost. FocusReID not only exceeds their accuracy but also remains far more efficient.

ConvNeXt shows strong performance (91.69% R1, 88.01% mAP), yet with the highest MACs (10.03G) and GPU usage (5653.01 MB) among all CNN-based models, making it unsuitable for real-time deployment. Similarly, HRNet delivers good accuracy (92.07% R1) but suffers from high inference time (2.78s). EfficientNet-B4 is lightweight (0.05G MACs) but lags in performance with only 90.80% R1 and 87.43% mAP.

Furthermore, we evaluated our FocusReID on transformer-based models, particularly the Swin Transformers, which are increasingly relevant in vision tasks. The Swin Transformer Small (SwinTrans_S) and Large (SwinTrans_L) illustrate the scalability of hierarchical transformers for PRiD.

SwinTrans_S achieves 91.71% R1 and 87.76% mAP with 28.36M parameters, but its higher inference time (2.09s) and GPU usage (3880.06 MB) limit its efficiency. SwinTrans_L improves performance (93.17% R1, 90.52% mAP), but at the cost of extreme computational demands: 196.48M parameters, 34.09G MACs, and 13,557.95 MB GPU usage. In contrast, FocusReID achieves the best overall performance (95.71% R1, 91.95% mAP) while maintaining efficiency. With only 9.02M parameters and 1.89G MACs, it outperforms even large transformer models through its lightweight FA module that enhances identity-relevant features and reduces background noise.

FocusReID achieves a remarkable balance between accuracy and computational efficiency. It consistently surpasses more complex and heavier models across both CNN-based and transformer-based categories, positioning it as a practical solution for real-time PRiD in resource-constrained environments. Its well-balanced performance profile effectively addresses the demand for high-accuracy and real-time capabilities in real-world surveillance applications.

4.5. Heatmaps visualization

In this section, we analyze and compare the Grad-CAM heatmap visualizations of the DenseNet variants (D121, D169, D161, D201) and the proposed FocusReID, as illustrated in Fig. 7. Compared to the baseline DenseNet models, FocusReID exhibits a more focused attention mechanism, concentrating on discriminative regions such as key body parts and accessories while effectively suppressing irrelevant background noise. In contrast, the DenseNet variants display scattered or less precise attention patterns, often activating on non-informative areas.

FocusReID heatmaps show stronger focus on key semantic regions, highlighting improved identity feature extraction. This is enabled by the FA module, which combines channel and spatial attention in D121, validating our dynamic attention strategy for enhanced discriminability in re-identification.

Table 6
Parametric and Performance comparison of backbone architectures on Market-1501.

Backbone	Parms (M)	MACs(G)	Time(s)	GPU(MB)	R1 (%)	mAP (%)
D121	8.89	1.89	0.97	2713.25	91.11%	84.52%
D201	21.38	2.87	1.82	4322.11	91.95%	88.87%
D169	15.39	2.24	1.32	3331.40	92.01%	88.66%
D161	30.20	5.12	2.56	5210.12	92.61%	89.53%
ResNet50	26.99	2.70	0.97	2105.93	90.32%	86.68%
ResNet101	45.98	5.14	1.30	3086.26	91.63%	88.35%
ResNet152	61.63	7.58	1.46	4278.18	91.89%	88.44%
ConvNeXt	88.48	10.03	2.70	5653.01	91.69%	88.01%
HRNet	20.69	2.84	2.78	2774.06	92.07%	88.22%
Effi-B4	18.85	0.05	1.15	5058.89	90.80%	87.43%
SwinTrans_S	28.36	4.35	2.09	3880.06	91.71%	87.76%
SwinTrans_L	196.48	34.09	6.89	13,557.95	93.17%	90.52%
FocusReID	9.02	1.89	1.01	2715.12	95.71%	91.95%

Note: The best results are highlighted in bold, while the second-best results are underlined.

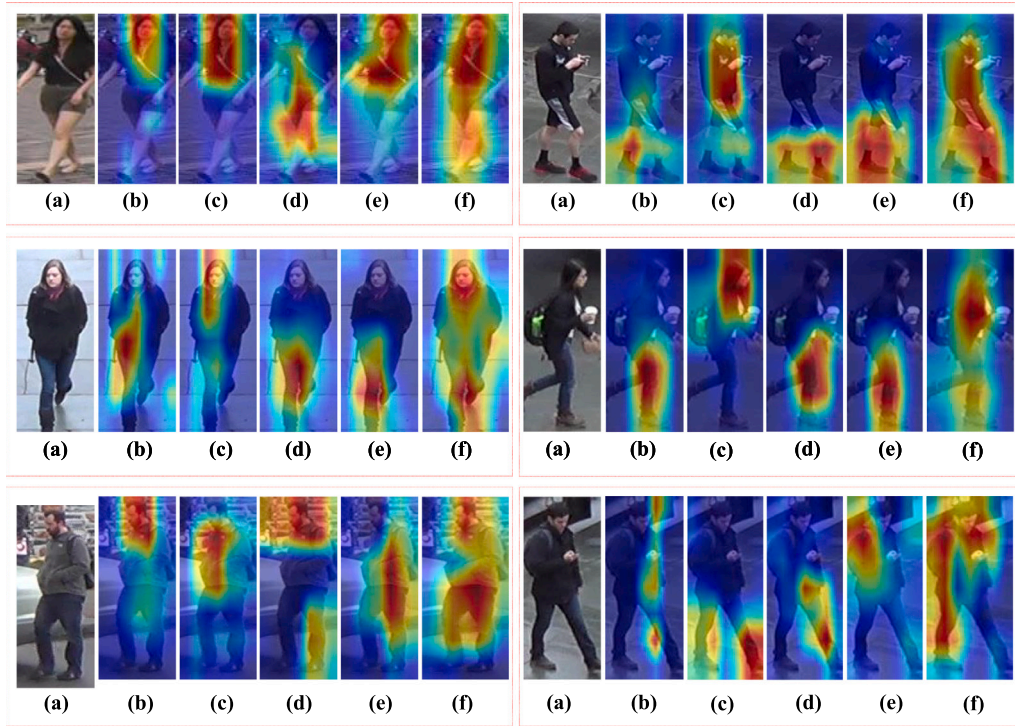


Fig. 7. Visualization of Grad-CAM heatmaps for DenseNet backbones (D121 baseline, D201, D169, D161) compared with FocusReID. (a) Input image, (b) D121, (c) D201, (d) D169, (e) D161, (f) FocusReID.

5. Ablation work

This section presents ablation studies validating the novelty of FocusReID for PReID. It examines the FA module, combining CA and SA, and the effects of hyperparameters like RE, LR-warmup, and learning strategies. The impact of different loss functions is also evaluated, with the CL+TL combination proving most effective. Integrated with the D121 backbone, these enhancements yield significant improvements on the Market dataset, consistently surpassing SOTA models. Key metrics are highlighted for clarity.

5.1. Impact of fusion attention

The integration of the FA module improves the performance of the proposed PReID framework, particularly in terms of R1 accuracy and mAP. The results presented in Fig. 8 illustrate these gains. Specifically, incorporating the FA module into the D121 backbone increases R1 accuracy by 4.60%, Rank-5 accuracy by 2.31%, and Rank-10 accuracy by 1.84%. In addition, mAP improves by 7.43%, suggesting that

the FA module contributes to more effective discriminative feature representation.

The effect of the FA module is further examined through controlled comparisons under the same experimental setting. As shown in Fig. 8, integrating FA into the D121 backbone consistently improves all evaluation metrics. This trend is also supported by the placement ablation in Table 7, where progressively introducing FA across multiple dense blocks yields stronger results than more limited insertion strategies. In addition, the stepwise ablation results in Table 8 provide complementary evidence by separating the effect of standard training strategies from that of the integrated FA module on the D121 backbone. Together, these results suggest that the gains observed in FocusReID are closely related to the proposed multi-stage attention integration.

5.2. Impact of FA module placement

We conduct an ablation study to evaluate the impact of FA placement across different dense blocks of D121. As summarized in Table 7, we progressively introduce FA modules from deeper to earlier layers,

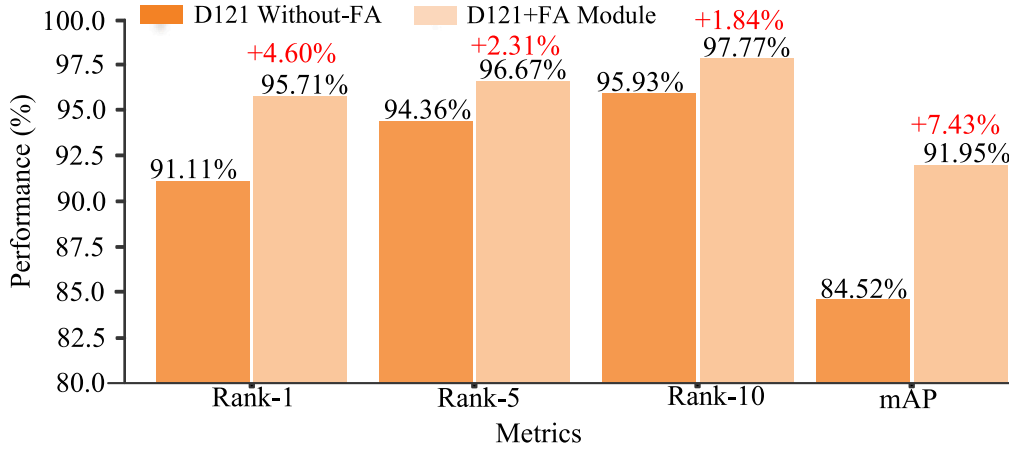


Fig. 8. Performance comparison of D121 with and without FA.

Table 7

Ablation study of FA module placement across D121 dense blocks.

Variant	D1	D2	D3	D4	R1 (%)	mAP (%)
D121 (baseline)	×	×	×	×	91.11%	84.52%
D121 + FA _{D4}	×	×	×	✓	92.22%	88.14%
D121 + FA _{D3,D4}	×	×	✓	✓	93.76%	90.68%
D121 + FA _{D2,D3,D4}	×	✓	✓	✓	93.91%	91.29%
FocusReID (FA _{D1,D2,D3,D4})	✓	✓	✓	✓	95.71%	91.95%

starting from the final dense block (D4) and extending to all dense blocks (D1–D4).

The results show that incorporating FA at the final stage (D4) improves the baseline from 91.11% to 92.22% in R1 and from 84.52% to 88.14% in mAP. Extending FA to D3 and D4 further increases performance to 93.76% R1 and 90.68% mAP, while adding FA to D2–D4 yields 93.91% R1 and 91.29% mAP. The full configuration achieves the strongest overall results among the tested placement settings, reaching 95.71% R1 and 91.95% mAP. This trend is also consistent with Table 6, where FocusReID shows a favorable accuracy–complexity trade-off relative to heavier models such as SwinTrans_L, achieving 95.71% / 91.95% with substantially lower complexity (9.02M parameters, 1.89G MACs) than 93.17% / 90.52% with 196.48M parameters and 34.09G MACs. These results suggest that the gains are progressively accumulated as FA is introduced across multiple stages, rather than arising from a single insertion point. This supports the effectiveness of multi-stage feature refinement during feature propagation and provides empirical justification for the proposed design.

5.3. Impact of training strategies and FA

To separate the effect of standard training strategies from that of the integrated FA module, we perform a stepwise ablation on the D121 backbone. We first evaluate the baseline model, then add the commonly used training strategies (Random Erasing, learning warm-up, and CL+TL loss), and finally introduce the FA module under the same training conditions.

As shown in Table 8, standard training strategies provide only moderate gains over the D121 baseline, improving R1 accuracy from 91.11% to 92.43% and mAP from 84.52% to 88.47%. In contrast, when the FA module is introduced under the same training configuration, performance increases significantly to 95.71% in R1 accuracy and 91.95% in mAP. Since the training setup remains identical between the second and third configurations, the additional improvement can be directly attributed to the FA module. Notably, the gain introduced by FA is substantially larger than that obtained from standard training strategies, demonstrating that the performance improvement is not merely due to their combination but is primarily driven by the proposed hierarchical attention integration.

5.4. Evaluating attention modules

This section evaluates the impact of different attention mechanisms integrated into the D121 backbone for PReID, using R1 accuracy and mAP as primary performance metrics. The proposed FA module, which combines CA and SA, achieves the highest performance. Specifically, FA improves R1 accuracy by 2.00% and mAP by 1.40% compared to Non-Local Attention (NLA), and it also outperforms Mix-Channel Attention (MCA), which provides only a marginal mAP gain of 0.14% while reducing R1 by 0.51%. Further analysis of individual attention types reveals that CA yields 91.98% R1 and 89.11% mAP, indicating its strength in emphasizing informative channels, albeit with limited spatial awareness. SA shows slightly better performance with 92.97% R1 and 89.92% mAP, highlighting the importance of spatial localization cues in PReID. The FA module, by fusing both CA and SA, consistently surpasses the individual components, demonstrating the effectiveness of joint spatial-channel refinement. As illustrated in Fig. 9, the FA module achieves 95.71% R1 and 91.95% mAP, confirming the benefit of adaptive, multi-level attention in enhancing discriminative feature learning and retrieval performance in PReID.

5.5. Losses & hyperparameter tuning

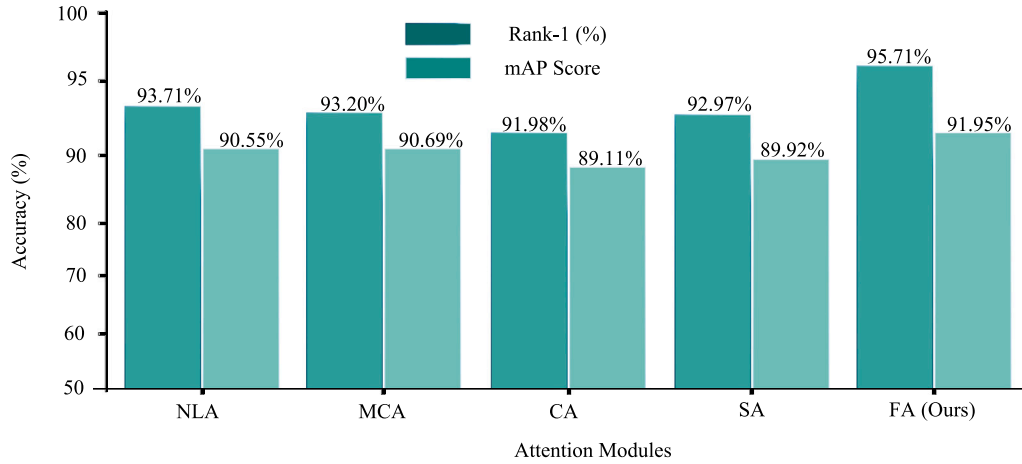
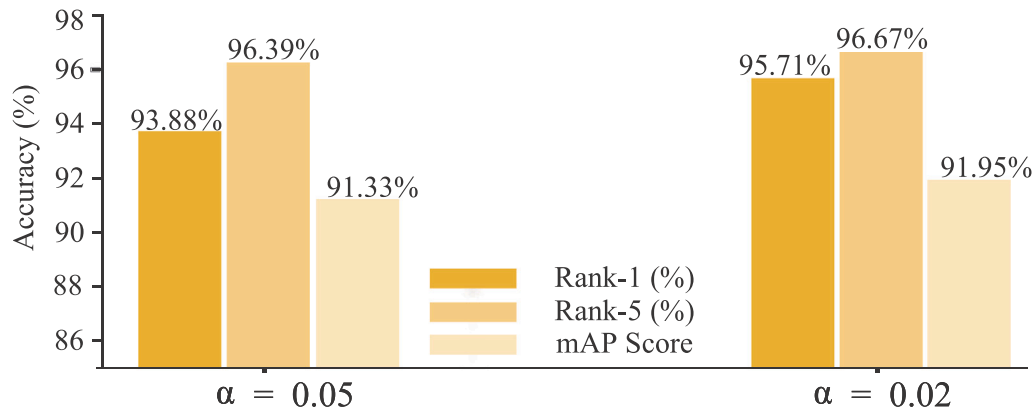
An ablation study was performed to refine critical hyperparameters for the FocusReID model in PReID tasks. Parameters such as batch size, learning rate, R-E probability, and W-up duration were systematically tuned. The best results were achieved using a batch size of 32, a learning rate of $\alpha = 0.02$, a R-E probability of 0.5, and a W-up phase lasting 5 epochs. Among the loss functions evaluated, the combination of CL and TL exhibited superior performance on the Market datasets. This combination outperforms other methods, achieving the highest R1 accuracy of 95.71% and mAP of 91.95%, indicating enhanced feature discrimination. While CTL achieved the highest R5 accuracy of 97.15%, its R1 performance is lower than that of the CL + TL combination. In contrast, TL showed impressive results with 94.36% R1 and 91.44% mAP. On the other hand, AFL and CFL demonstrate significantly poorer results. These findings highlight the efficacy of combining loss functions to achieve optimized performance in tasks such as PReID (see Table 9).

Furthermore, we analyzed the effect of reducing the α from 0.05 to 0.02, with other settings unchanged, enhanced performance (see Fig. 10). R1 increased from 93.88% to 95.71%, mAP from 91.33% to 91.95%, and R5 slightly increase. The lower α ensured stable convergence and better generalization, making $\alpha = 0.02$ the optimal choice.

Table 8

Separate contribution of training strategies and the FA module on Market-1501 using the D121 backbone.

Config.	R-E	LW-up	CL+TL	B=32	$\alpha=0.02$	FA	R1	mAP
D121 Baseline	✗	✗	✗	✓	✓	✗	91.11%	84.52%
D121 + Tricks	✓	✓	✓	✓	✓	✗	92.43%	88.47%
FocusReID	✓	✓	✓	✓	✓	✓	95.71%	91.95%

**Fig. 9.** R1 and mAP (%) comparison of attention modules on D121.**Fig. 10.** Effect of learning rate (α) on CL + TL performance: 0.05 vs 0.02.**Table 9**

Performance comparison across different loss functions.

Loss Function	R1 (%)	R5 (%)	mAP (%)
CL	93.94%	96.88%	90.87%
CEL	93.76%	96.79%	91.10%
AFL	86.70%	92.43%	78.03%
CFL	89.25%	93.82%	82.51%
SFL	94.39%	96.76%	90.98%
CTL	93.91%	97.15%	91.46%
TL	94.36%	97.21%	91.44%
CL + TL	95.71%	96.67%	91.95%

Note: The best results are in **bold**, and the second-best are underlined.

5.6. Impact of hyperparameter

Quantitative analysis of hyperparameter variations across diverse backbone architectures for PReID on the Market-1501 dataset, as detailed in Table 10, reveals significant performance improvements with the incorporation of RE, α , LW-up, CL, and TL. The analysis shows substantial gains in both R1 accuracy and mAP. Specifically, the D121 Baseline saw a 1.32% increase in R1 92.43% and a 3.95% improvement

in mAP 88.47%. ResNet50 achieved a 2.11% increase in R1 92.43% and a 1.79% rise in mAP 88.47%. More modest gains were observed in ResNet50-IBN 0.44% R1, 1.27% mAP, EfficientNet-B4 1.63% R1, 1.04% mAP, and HRNet 0.36% R1, 0.25% mAP. Notably, FocusReID achieved the highest performance, with 95.71% R1 and 91.95% mAP, surpassing HRNet by 3.64% in R1 and 3.48% in mAP. These findings emphasize the effectiveness of the applied techniques in enhancing feature discrimination and retrieval accuracy across various backbone models.

5.7. Case study

This section evaluates the robustness of FocusReID against occlusions, resolution variations, retrieval examples, and failure cases.

Case 1: Occlusion robustness

The occlusion robustness of FocusReID was evaluated using the Occluded-DukeMTMC dataset. Results (Fig. 12) revealed a marked performance difference between the baseline without inclusion of FA (WOFA) and proposed FocusReID. WOFA achieved 56.31% R1 and

Table 10
Impact of hyperparameters on different backbones for PReID (Market-1501).

Backbone	R-E	LW-up	CL+TL	B=32	$\alpha = 0.02$	R1 (%)	mAP (%)
D121 (base)	✗	✗	✗	✓	✓	91.11%	84.52%
D121 (+all)	✓	✓	✓	✓	✓	92.43%	88.47%
RN50	✓	✓	✓	✓	✓	90.32%	86.68%
RN50-IBN	✓	✓	✓	✓	✓	91.99%	87.20%
EN-B4	✓	✓	✓	✓	✓	90.80%	87.43%
HRN	✓	✓	✓	✓	✓	92.07%	88.22%
CNX	✓	✓	✓	✓	✓	91.69%	88.01%
FocusReID	✓	✓	✓	✓	✓	95.71%	91.95%

Note: D121 = DenseNet-121; RN50 = ResNet-50; RN50-IBN = ResNet-50 with Instance-Batch Normalization; EN-B4 = EfficientNet-B4; HRN = HRNet; CNX = ConvNeXt; R-E = Random Erasing; LW-up = learning warm-up.



Fig. 11. Comparative retrieval results on Market-1501: green borders show correct matches, and red indicates incorrect matches. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

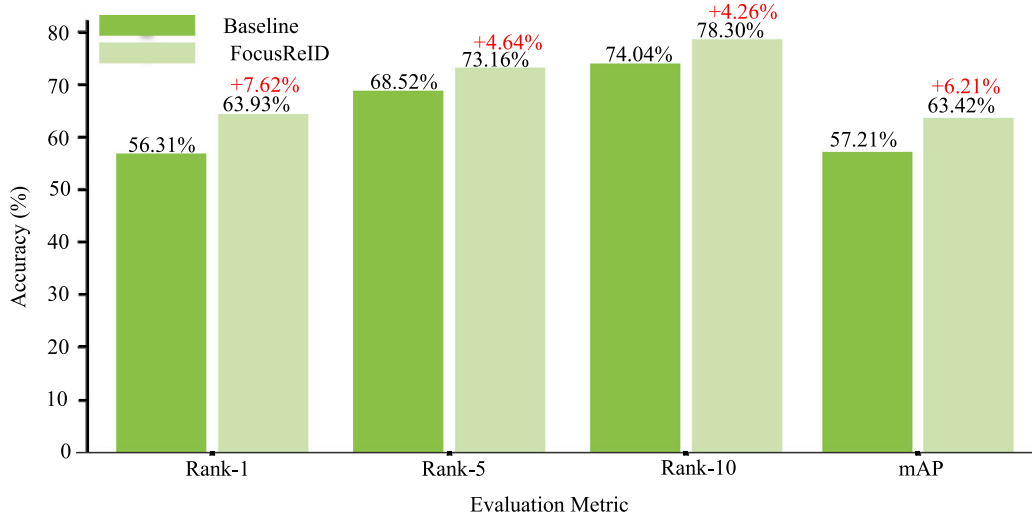


Fig. 12. Occlusion robustness: Baseline vs. FocusReID.

57.21% mAP, indicating sensitivity to occlusions. FocusReID, incorporating FA, significantly strengthens performance, achieving 63.93% R1 and 63.42% mAP. Gains were also observed in higher-order rankings R5: 73.16%, R10: 78.30%, demonstrating FA's efficacy in extracting discriminative features under occlusion, crucial for real-world applications.

Case 2: Resolution sensitivity

To assess resolution sensitivity, experiments were conducted on the Market-1501 dataset, reducing the original resolution of 64×128 to

32×64 (half) and 16×32 (quarter). As shown in Table 11, FocusReID outperformed the baseline at all resolutions. At the original resolution, FocusReID achieved 95.71% R1 and 91.95% mAP, surpassing the baseline by 3.28% and 3.48%. At 32×64 , FocusReID improved by 1.69% R1 and 2.76% mAP. Even at the degraded resolution of 16×32 , FocusReID maintained a significant advantage, achieving 79.82% R1 and 69.93% mAP, surpassing the baseline by 2.86% and 5.38%, respectively. These results demonstrate FocusReID's robustness to resolution degradation and its ability to maintain performance under low resolution, supporting real-time use cases.

Table 11

Resolution performance comparison between the baseline and FocusReID on Market-1501.

Resolution (W×H)	Baseline		FocusReID	
	R1 (%)	mAP (%)	R1 (%)	mAP (%)
64 × 128	92.43%	88.47%	95.71%	91.95%
32 × 64	90.29%	86.47%	91.98%	89.23%
16 × 32	76.96%	64.55%	79.82%	69.93%

Note: All values are reported as percentages. The best results are highlighted in **bold**.

5.8. Case 3: Retrieval & failure case

We evaluated the top-10 retrieval performance of FocusReID in comparison to the baseline D121 model on the Market1501 dataset. As shown in Fig. 11(a), the baseline model retrieves 5 out of 10 correct matches, indicating limited robustness in the presence of common PreID challenges such as occlusion and background clutter. In contrast, Fig. 11(b) presents the results of FocusReID, which correctly retrieves 8 out of 10 matches (indicated by green borders), demonstrating a substantial improvement in retrieval accuracy. This enhanced performance underscores the effectiveness of the proposed attention mechanisms in refining feature representations and improving robustness against visual variations.

The analysis highlights three baseline limitations: (1) sensitivity to lighting variations, (2) dependence on viewpoint consistency, and (3) susceptibility to accessory interference. Although FocusReID outperforms overall, it occasionally fails under extreme pose variations, indicating a need to enhance robustness against combined pose and viewpoint challenges. In addition, the cross-dataset results indicate that domain shift remains a limitation, suggesting the need for further improvement in cross-domain robustness.

6. Conclusion

This paper presented FocusReID, a lightweight PreID framework designed to improve robustness under occlusion, appearance variation, and background clutter. By integrating the FA module into the D121 backbone, the proposed framework progressively refines hierarchical feature representations and enhances identity-discriminative cues while suppressing irrelevant background information. FocusReID achieves a favorable balance between accuracy and efficiency, requiring only 9.02M parameters, 1.89G MACs, and 1.01s inference time. In addition, the use of Random Erasing, learning warm-up, and the hybrid Circle Loss and Triplet Loss supports stable optimization and effective retrieval-oriented feature learning. Extensive experiments on benchmark datasets show that FocusReID achieves competitive performance, reaching Rank-1 accuracies of 95.71% on Market-1501 and 91.10% on DukeMTMC-reID, with strong results also observed on Occluded-Duke. These findings indicate that FocusReID provides an effective accuracy–efficiency tradeoff for practical PreID applications. We acknowledge that the novelty of the proposed method lies mainly in the systematic architectural integration of attention for progressive refinement, rather than in introducing a fundamentally new attention formulation. In future work, we will further investigate cross-backbone validation and more advanced attention designs to enhance generalization and representation capability.

CRedit authorship contribution statement

Tariq Ali Arain: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Pengcheng Zhang:** Supervision, Conceptualization. **Sehrish Mazhar:** Writing – review & editing, Visualization. **Qing Meng:** Supervision, Conceptualization. **Imran Ali Soomro:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data are publicly available and cited in the manuscript.

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